Kuta Software - Infinite Precalculus

Name________________________________

Logarithmic Equations

Date________________ Period____

Solve each equation.

1) $\log (3x - 9) = \log (2x + 6)$

2) $\log (-4n + 7) = \log 3n$

3) $\log n = \log 12$

4) $\log (5x - 7) = \log (3x - 1)$

5) $1 + \log_5 -9b = 4$

6) $-7\log_4 -10r = -14$

7) $4\log_{11} (r + 8) = 8$

8) $\log_3 (x + 1) - 5 = -5$

9) $\log_{18} (3k^2 - 5k) = \log_{18} (-6 + 2k^2)$

10) $\log_{14} (6v - 1) = \log_{14} (v^2 - 17)$

11) $\log_{19} (7 - 3r^2) = \log_{19} (-2r^2 - 6r)$

12) $\log_{14} (-32 - 3n) = \log_{14} (n^2 + 9n)$